Labor Markets, Distribution and the Political Economy of International Trade

ECON 404: Cal Poly

Motivation

- Common belief among economists...
  - Free trade or freer trade is "good"
- But... trade has distributional implications
  - HO model, Specific factors model, etc.
  - Why is free or freer trade good?
    - Gains to winners $>$ losses to losers
- And... these distributional implications have consequences for
  - Political viability of trade related policy
  - Political outcomes more generally
    - Composition of elected politicians by party and ideology

Distributional implications of trade

- Strong theoretical predictions about distributional implications
  - HO: rising wage inequality between skilled and unskilled workers
  - Specific factors model: contracting sector specific factor hurt
- Empirical evidence for strong distributional implications?
  - Until recently, relatively little
  - Recently, lots of strong evidence

Outlets for trade-related distributional tensions

1. Political resistance to trade reform by voters
   - Affects politician's legislative voting behavior
     - Role for compensating "losers" of trade reform?
2. Voters favor "anti-trade" politicians
   - Affects composition of legislature
     - Party lines?
     - Ideological lines within party?
   - No time to cover fully today...

Outline

1. Distributional implications: empirical evidence
   - Autor, Dorn & Hanson (2013, AER)
   - Pierce & Schott (2016, AER)
   - McLaren & Hakobyan (2016, RESTAT)
2. Outlets for trade-related distributional tensions: empirical evidence
   - Reminder
   - Lake & Millimet (JIE, 2016)
3. Conclusion

Motivation and research question

- Motivation
  - Rapid rise of China
    - 1990-2007: massive $\uparrow$ US imports from China
- Research question
  - How does this affect US labor market outcomes?
    - Manufacturing employment, unemployment, labor force participation, wages

Econometrics: thought experiment

- Look at change in labor market outcomes 1990-2007
- Control group
  - US *locations* whose workers don't compete with China
- Treatment group
  - US *locations* whose workers compete with China

Need local measures of rising import penetration

Econometrics: problem

- Two reasons for rising US imports from China

1. US product demand shocks
   - EX. US consumers start buying more clothes
     - Some produced domestically $\Rightarrow\uparrow$ US employment
     - Some produced by China $\Rightarrow\uparrow$ Chinese IM
   - Hard to estimate ceteris paribus effect of $\uparrow$ Chines IM
2. Chinese export supply shocks
   - Free market reforms, productivity growth, etc.

- Ideally want to know effect of #2

Need local measures of rising import penetration due to Chinese export supply shocks

Econometrics: Chinese export supply shocks

- How do we identify Chinese export supply shocks?
- Suppose China exports $\uparrow$ because of export supply shocks
- Then, expect $\uparrow$ Chinese exports to *other* countries too

Effect on manufacturing employment share

Table 3. Imports from China and Change of Manufacturing Employment in Commuting Zones, 1990-2007: 2SLS Estimates.

Dependent Var: 10 x Annual Change in Manufacturing Emp/Working Age Pop (in %pts)

	I. 1990-2007 Stacked First Differences					
	(1)	(2)	(3)	(4)	(5)	(6)
(Δ Imports from China to US)/Worker	-0.746 **	-0.610 **	-0.538 **	-0.508 **	-0.562 **	-0.596 **
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
Percentage of employment in manufacturing$_{-1}$		-0.035	-0.052 **	-0.061 **	-0.056 **	-0.040 **
		(0.022)	(0.020)	(0.017)	(0.016)	(0.013)
Percentage of college-educated population$_{-1}$				-0.008		0.013
				(0.016)		(0.012)
Percentage of foreign-born population$_{-1}$				-0.007		0.030 **
				(0.008)		(0.011)
Percentage of employment among women$_{-1}$				-0.054 *		-0.006
				(0.025)		(0.024)
Percentage of employment in routine occupations$_{-1}$					-0.230 **	-0.245 **
					(0.063)	(0.064)
Average offshorability index of occupations$_{-1}$					0.244	-0.059
					(0.252)	(0.237)
Census division dummies	No	No	Yes	Yes	Yes	Yes
	II. 2SLS First Stage Estimates					
(Δ Imports from China to OTH)/Worker	0.792 **	0.664 **	0.652 **	0.635 **	0.638 **	0.631 **
	(0.079)	(0.086)	(0.090)	(0.090)	(0.087)	(0.087)
R^2	0.54	0.57	0.58	0.58	0.58	0.58

Is this a big effect?

Yes...

- US manufacturing employment share
  - B/w 1990 and 2007: $\downarrow$ 4.18% points
  - In 1990, 12.6%. So $\approx$ 33% of 1990 level
- About 22.5% of the 4.18% pt decline is due to the Chinese export supply shock

Do people move away?

Table 4. Imports from China and Change of Working Age Population in Commuting Zones, 1990-2
2SLS Estimates.
Dependent Variables: 10-Year Equivalent Changes in Log Population Counts (in log pts)

	I. By Education Level			II. By Age Group		
	All	College	Non-College	Age 16-34	Age 35-49	A
	(1)	(2)	(3)	(4)	(5)	
	A. No Census Division Dummies or Other Controls					
(Δ Imports from China to US)/Worker	-1.031 *	-0.360	-1.097 *	-1.299	-0.615	
	(0.503)	(0.660)	(0.488)	(0.826)	(0.572)	
R^2	.	0.03	0.00	0.17	0.59	
	B. Controlling for Census Division Dummies					
(Δ Imports from China to US)/Worker	-0.355	0.147	-0.240	-0.408	-0.045	
	(0.513)	(0.619)	(0.519)	(0.953)	(0.474)	
R^2	0.36	0.29	0.45	0.42	0.68	
	C. Full Controls					
(Δ Imports from China to US)/Worker	-0.050	-0.026	-0.047	-0.138	0.367	
	(0.746)	(0.685)	(0.823)	(1.190)	(0.560)	
R^2	0.42	0.35	0.52	0.44	0.75	

What happens to displaced workers?

Table 5. Imports from China and Employment Status of Working Age Population within Commuting Zones, 1990-2007: 2SLS Estimates.
Dep Vars: 10-Year Equivalent Changes in Log Population Counts and Population Shares by Employment Status

	Mfg Emp (1)		Non-Mfg Emp (2)		Unemp (3)		NILF (4)		SSDI Receipt (5)	
					A. 100 × Log Change in Population Counts					
(Δ Imports from China to US)/Worker	-4.231 (1.047)	**	-0.274 (0.651)		4.921 (1.128)	**	2.058 (1.080)	~	1.466 (0.557)	**
					B. Change in Population Shares					
					All Education Levels					
(Δ Imports from China to US)/Worker	-0.596 (0.099)	**	-0.178 (0.137)		0.221 (0.058)	**	0.553 (0.150)	**	0.076 (0.028)	**
					College Education					
(Δ Imports from China to US)/Worker	-0.592 (0.125)	**	0.168 (0.122)		0.119 (0.039)	**	0.304 (0.113)	**	.	
					No College Education					
(Δ Imports from China to US)/Worker	-0.581 (0.095)	**	-0.531 (0.203)	**	0.282 (0.085)	**	0.831 (0.211)	**	.	

Effects on non-manufacturing workers

Table 7. Comparing Employment and Wage Changes in Manufacturing and outside Manufacturing, 1990-2007: 2SLS Estimates.

Dep Vars: 10-Year Equiv. Changes in Log Workers (in Log Pts) and Avg Log Weekly Wages (in %)

	I. Manufacturing Sector						II. Non-Manufacturing					
	All Workers (1)		College (2)		Non-College (3)		All Workers (4)		College (5)		Non-College (6)	
	A. Log Change in Number of Workers											
(Δ Imports from China to US)/Worker	-4.231	**	-3.992	**	-4.493	**	-0.274		0.291		-1.037	
	(1.047)		(1.181)		(1.243)		(0.651)		(0.590)		(0.764)	
R^2	0.31		0.30		0.34		0.35		0.29		0.53	
	B. Change in Average Log Wage											
(Δ Imports from China to US)/Worker	0.150		0.458		-0.101		-0.761	**	-0.743	*	-0.822	**
	(0.482)		(0.340)		(0.369)		(0.260)		(0.297)		(0.246)	
R^2	0.22		0.21		0.33		0.60		0.54		0.51	

Notes: N=1444 (722 commuting zones x 2 time periods). All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period commuting zone share of national population. ~ $p \leq 0.10$, * $p \leq 0.05$, ** $p \leq 0.01$.

Summary of main results

1. Chinese export supply shocks explain about 22% of US manuf emp decline b/w 1990-2007
2. Workers do not migrate away from heavily affected locations
3. Non-manuf sector does not expand to absorb displaced manuf workers. Rather they go into unemployment or leave the labor force.
    - Especially true if not college educated
4. But, increased competition for non-manuf sector job bids down non-manuf wages
5. Distributional transfers increase (disability, federal income assistance)

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Motivation and research question

Motivation

- WTO must treat other WTO members equally
  - Most Favored Nation (MFN) principle
  - Exception: Preferential Trade Agreements (e.g. NAFTA)
- US policy on non-WTO members
  - Grant "normal trade relations (NTR)" $\Rightarrow$ impose MFN tariffs
  - Don't grant NTR $\Rightarrow$ impose *very high* non-NTR tariffs
    - Non-NTR tariffs set in 1930
    - So, *unrelated* to current economic/political conditions

Motivation and research question

- Pre-2001, US annually decided whether to grant NTR
  - From 1980, US granted NTR status to China every year
  - But... politically controversial during 1990s
- China entered WTO in 2001
- WTO entry $\Rightarrow$ permanent NTR (PNTR)
  - Effectively, US liberalizes tariff policy on China
  - Potential threat of very high non-NTR tariffs
    - Depressed US imports from China
  - Threat now gone

Research question

- Effect of PNTR on US mauf employment?

Econometrics: thought experiment

- Look at change in industry employment pre and post PNTR
- Control group
  - US industries where non-NTR tariff $=$ NTR tariff
- Treatment group
  - US industries where non-NTR tariff $>$ NTR tariff

Need industry specific measure of NTR gap

Main result

- Use annual industry-level data 1990-2007
- Finds the elimination of NTRgap in all industries $\Rightarrow\downarrow$ aggregate manufacture employment by 15%
  - relative to the % change in manufacture employment in hypothetical situation where NTRgap didn't exist to begin with

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Motivation and research question

Motivation

- NAFTA very controversial in US
  - 1994: NAFTA comes into force
  - 2000: US has eliminated most tariffs on Mexico

Research question

- Effects of falling NAFTA tariffs on US wages?
  - When tariff for *own* industry falls
  - When tariffs for important *local* industries fall
  - Heterogeneity by worker education level?

Econometrics: thought experiment

- Look at change in worker wages 1990-2000
- Control group (location tariffs)
  - US workers in *locations* with many workers in industries where
    - "small" fall in tariffs on Mexico or
    - Mexico doesn't have a comparative advantage
- Treatment group (location tariffs)
  - US workers in *locations* with many workers in industries where
    - "big" fall in tariffs on Mexico or
    - Mexico has a comparative advantage

Need local measure of "local tariff exposure"

Econometrics: thought experiment

- Look at change in worker wages 1990-2000
- Control group (industry tariffs)
  - US workers in *industries* where
    - "small" fall in tariffs on Mexico or
    - Mexico doesn't have a comparative advantage
- Treatment group (industry tariffs)
  - US workers in *industries* where
    - "big" fall in tariffs on Mexico or
    - Mexico has a comparative advantage
- Really 8 thought experiments...
  - 1 on this slide and 1 on last slide, but do this for each of 4 educational groups

Interpreting the results

- High school dropouts (often blue collarworkers) hit hard in vulnerable locations & industries
- *Relative* to wage growth of HS dropout in unaffected industry and location
  - NAFTA reduces wage growth of HS dropout by...
  - 16% pts in most vulnerable industry but unaffected location
  - 10% pts in unaffected industry but most vulnerable location
  - 25% pts in most vulnerable industry and most vulnerable location
- High school grads and some college: somewhat weaker effects
- College grads: no effects

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Motivation and research question

Motivation

- Trade has strong distributional implications
  - Politician rhetoric echoes voter anxieties
    - Trump, Sanders, Clinton
- Likely generates political resistance to trade reform
  - US Congressman vote on Free Trade Agreements (FTAs)
    - 13 in total (e.g. NAFTA, Colombia, etc.)

Motivation and research question

- Will a Congressman vote in favor of an FTA?
  - Less likely if many losers in their Congressional District (CD)
  - Perhaps more likely if losers (partly) compensated
- US has Trade Adjustment Assistance (TAA) program
  - Trade displaced workers get income support and training
  - TAA petition needs approval from Dept. of Labor

Research question

- Are Congressman more likely to vote in favor of FTAs when expected trade related redistribution higher in their CD?

Econometrics: thought experiment

- Does a given Congressman change their FTA voting behavior when expected trade related redistribution (R) higher in their CD?
- Control group
    - Congressman voting on FTAs when R low in their CD
- Treatment group
    - Congressman voting on FTAs when R high in their CD

Need local measures of expected trade related redistribution

Econometrics: local expected trade related redistribution

- Notation
  - c: location (435 Congressional Districts)
  - s: state (50 states)
  - j: industry ($\approx$ 400 4-digit SIC industries)
  - t: year (annual, 2001-2011)
  - L: employment
- Expected trade related redistribution in location c time t

$$R_{ct} = UI_{st} \times P_{ct}$$

  - UI_{st}: unemployment insurance in state s at time t
  - P_{ct}: prob of trade displaced worker approved for TAA

$$P_{ct} = \sum_j \frac{L_{cj,2000}}{L_{c,2000}} P_{jt}$$

Econometrics: local expected trade related redistribution

$$R_{ct} = UI_{st} \times P_{ct}$$

$$P_{ct} = \sum_{j} \frac{L_{cj,2000}}{L_{c,2000}} P_{jt}$$

- When is R_{ct} large?
  - TAA income support generous in state of location c, and...
  - Workers in *location c* concentrated in *industries* where trade-displaced workers very likely to be approved for TAA

Econometrics: regression

- Notation
  - v: vote ($= 1$ if pro-FTA)
  - i: Congressman (435 in each Congress)
  - b: FTA bill (11 in sample)
- Main specification

$$v_{ibt} = \theta R_{ct} + \beta\tau_{cb} + \ldots$$

- θ: effect of expected trade related redist on pro-FTA vote
- τ_{cb}: local tariff on FTA partner b

Econometrics: results

$$v_{ibt} = \theta R_{ct} + \beta \tau_{cb} + \dots$$

- $\theta > 0$: more likely to vote in favor of FTA when R_{ct} higher
- How much more likely?
  - Effect of 1 stdev $\downarrow$ in R about 20-30% weaker than 1 stdev $\uparrow$ in τ_{cb}
  - 1 stdev $\downarrow$ in R for all locations would've blocked two FTAs
- But, very heterogeneous effects
  - Effect of 1 stdev $\uparrow$ in R much larger when
    - high τ_{cb}
    - high re-election risk
  - "Vulnerable" politicians very responsive to TAA

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Conclusion

- Trade has strong distributional implications for particular groups of workers and locations
- Outlets for distributional tensions
  - Political resistance to trade reform
    - Trade related redistribution mitigates this resistance